# How Lag-Llama makes a forecast

Rasul et al. · 2.5M parameters · LLaMA-style decoder on lagged values · zero-shot

## 1. One series, with its dates

Lag-Llama reads only the target's own past. A ? is a value not known yet.

*past* *future*

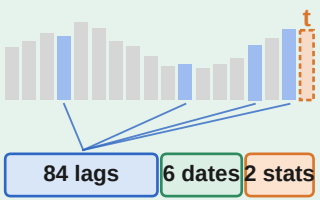
**Target** temperature

The dates matter: each step also carries six calendar features.

No covariates; each metal on its own.

## 2. A token built from lags

Values are scaled by the median and the interquartile range. The token for step t holds **84 earlier values**, up to 1,093 steps back.



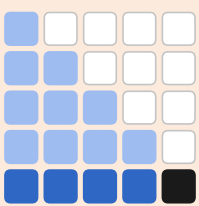

*one token for step t*

The two statistics record the median and spread used for scaling, so the level is not lost.

## 3. Eight causal blocks

The last 32 tokens go through a small LLaMA-style decoder. RMSNorm in front, shortcut around.

*earlier → later*



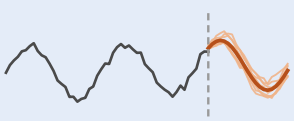
*earlier steps only*

**Self-attention:** causal, 9 heads of 16 (width 144), RoPE

**Feed forward:** SwiGLU, 144 → 512 → 144

## 4. One step at a time

The last token gives a **Student-t distribution** for the next step. A value is sampled, appended, and the model runs again.



median

sample paths

100 paths; their median is the point forecast. A 90-day horizon takes 90 passes.

**One of the first open forecasting foundation models: tiny, and built on lags rather than patches.**

8 blocks · width 144 · 9 heads · 84 lags + 6 date features + 2 statistics per token · context of 32 tokens

In the notebooks: context of 32 tokens, linear RoPE scaling for longer horizons · each metal on its own · no covariates · 100 samples